

BiGGViz: A MATLAB App for Visualization of Genome-Scale BiGG-Based Metabolic Networks

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Abstract

Motivation: Genome-scale metabolic reconstructions have become essential tools in systems biology for studying cellular metabolism and supporting *in silico* experimentation. The Biochemical, Genetic, and Genomic (BiGG) knowledge base provides standardized, open-source metabolic models in multiple formats, including the Systems Biology Markup Language (SBML) and MAT formats, facilitating model reuse and integration with computational frameworks such as the COBRA Toolbox. However, visualization and exploratory analysis of large-scale metabolic networks remain challenging, particularly for users without extensive programming expertise. Accessible and interactive visualization tools are needed to improve interpretation, communication, and exploration of dense reaction and metabolite networks.

Results: We developed BiGGViz, a MATLAB-based application for interactive visualization and exploration of COBRA-compatible BiGG metabolic models. BiGGViz automatically constructs reaction-reaction and metabolite-metabolite graphs and provides an interactive graphical interface for network exploration. Users can switch between reaction- and metabolite-centric views, customize node appearance, apply **topology-aware** filtering, inspect local neighborhoods, highlight pathways with configurable hop-based expansion, and export visualizations or network structures for downstream analyses. Optional annotation tables support overlay of user-defined attributes and COBRA-derived analyses, including flux balance analysis (FBA) results, enabling flux-aware exploration of genome-scale metabolic networks. BiGGViz supports exploratory systems biology analyses, result communication, educational applications, and integration with downstream Cytoscape-based workflows. A scriptable wrapper, `exportBiGGViz`, is additionally provided to support reproducible and code-driven analysis pipelines.

Availability and Implementation: BiGGViz is implemented in MATLAB and is freely available at <https://github.com/mogystats/BiGGViz>. Example datasets and documentation are provided within the repository.

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Keywords: genome-scale metabolic models, COBRA Toolbox, metabolic network visualization, systems biology, SBML

Abbreviations: BiGG, Biochemical, Genetic, and Genomic; COBRA, Constraint-Based Reconstruction and Analysis; FBA, Flux Balance Analysis; FVA, Flux Variability Analysis; SBML, Systems Biology Markup Language

Introduction

Genome-scale metabolic reconstructions provide structured representations of cellular metabolism by organizing known metabolic reactions, metabolites, and biochemical relationships into stoichiometric networks. Within the **CO**nstraint-Based **R**econstruction and Analysis (**COBRA**) **T**oolbox framework, these reconstructions are transformed into computable models that support *in silico* investigation of metabolic behavior under physicochemical and environmental constraints (Orth et al., 2010; Schellenberger et al., 2011; Heirendt et al., 2019). Common analyses include flux balance analysis (FBA), flux variability analysis (FVA), reaction and gene knockout studies, and integration of transcriptomic or metabolomic data for generation

of context-specific models (Gudmundsson and Thiele, 2010; Becker and Palsson, 2008). Collectively, these approaches have become central tools in systems biology for studying metabolic capabilities across organisms, environmental conditions, and experimental perturbations.

The Biochemical, Genetic, and Genomic (BiGG) Models knowledge base provides a widely used repository of curated genome-scale metabolic reconstructions with standardized metabolite and reaction identifiers, rich annotations, and direct compatibility with COBRA-based workflows (King et al., 2016). In practice, BiGG-derived models are frequently loaded into the COBRA Toolbox for downstream analysis and simulation. However, interpretation of COBRA-derived outputs often requires

Table 1. Comparison of BiGGViz and related metabolic visualization tools.

Feature	BiGGViz	Escher	Cytoscape	MetExploreViz
Automatic network generation from SBML/MAT	Yes	Limited	Limited	Yes
Reaction-reaction networks	Yes	No	Yes	Yes
Metabolite-metabolite networks	Yes	Limited	Yes	Yes
Topology-aware filtering	Yes	Limited	Yes	Limited
Dynamic annotation overlays	Yes	Limited	Yes	Limited
FBA/FVA integration	Yes	Limited	Limited	Limited
Neighborhood exploration	Yes	Limited	Yes	Limited
Interactive HTML export	Yes	Yes	Limited	Yes
MATLAB workflow integration	Yes	No	No	No

understanding how predicted fluxes, blocked reactions, pathway perturbations, or context-specific subnetworks relate to the broader metabolic network structure. This becomes increasingly difficult at genome scale, where models may contain thousands of reactions and metabolites connected through dense and highly interconnected biochemical relationships.

Network visualization and exploratory analysis approaches can provide important biological context for interpreting these large-scale metabolic systems. General-purpose platforms such as Cytoscape support flexible graph visualization and network analysis workflows (Shannon et al., 2003), while Escher emphasizes curated pathway-based visualization of metabolic maps and pathway-level data integration (King et al., 2015). Additional tools, including BiGGR, SBEToolbox, MetaboNetworks, and MetExploreViz, support various aspects of metabolic modeling, pathway analysis, or network interrogation within distinct computational ecosystems (Gavai et al., 2015; Konganti et al., 2013; Posma et al., 2014; Chazalviel et al., 2018). These platforms collectively provide valuable capabilities for metabolic systems biology, although they frequently emphasize either pathway-centric visualization, external graph analytics, or specialized downstream analyses rather than direct exploratory interrogation of COBRA-compatible network topology within MATLAB-based workflows.

In addition to pathway-level interpretation, graph-derived network representations provide complementary perspectives on biochemical organization and metabolic connectivity within genome-scale reconstructions. Reaction-reaction and metabolite-metabolite networks enable investigation of topological organization, pathway cross-talk, local neighborhood structure, and highly connected hub metabolites within large metabolic systems. These representations additionally facilitate downstream graph-theoretic analyses, including estimation of centrality measures, community detection, subnetwork extraction, and topology-aware interpretation of perturbation responses. As genome-scale reconstructions continue to grow in size and complexity, exploratory network-centric workflows may provide important context for interpreting COBRA-derived analytical outputs and condition-specific metabolic behavior.

In practice, workflows involving COBRA-derived analyses often require transitions between multiple software environments for flux analysis, graph construction, annotation integration, and downstream visualization. Although Cytoscape provides a rich ecosystem for graph analytics and network-based interpretation, metabolic reconstructions typically require preprocessing and export into external graph formats prior to exploratory analysis.

Similarly, Escher provides biologically intuitive pathway visualizations but is oriented primarily toward curated pathway-map representations rather than direct interrogation of stoichiometric reconstructions across the whole network. MetExploreViz and related environments provide additional interactive capabilities for metabolic network interrogation, although functionality may remain constrained by specific workflows, external preprocessing requirements, or restricted support for generalized SBML- and MAT-driven network generation. As summarized in Table 1, few existing tools combine generalized SBML/MAT-driven network generation, reaction- and metabolite-centric exploration, dynamic analytical overlays, and topology-aware workflows within native MATLAB/COBRA environments. In Table 1, “Limited” indicates constrained or indirect functionality requiring curated workflows, preprocessing, or partial feature support.

BiGGViz is a MATLAB-based visualization framework for interactive exploration of BiGG-derived COBRA metabolic models. BiGGViz directly accepts COBRA-compatible model structures and derives reaction-reaction and metabolite-metabolite network projections from the underlying stoichiometric matrix. More specifically, BiGGViz interprets the stoichiometric matrix as an underlying bipartite reaction-metabolite graph from which reaction-reaction and metabolite-metabolite projections are derived. The framework supports reaction- and metabolite-centric visualization, configurable filtering strategies, node- and pathway-focused neighborhood expansion, and overlay of user-defined or COBRA-derived annotations, including FBA-derived flux information. In addition to the graphical application, BiGGViz includes a scriptable export interface that supports reproducible, code-driven analysis pipelines. By operating directly within the MATLAB/COBRA ecosystem, BiGGViz provides an integrated visualization layer that complements downstream systems biology analysis, exploratory network interrogation, graph-theoretic workflows, and scientific communication applications. The graphical interface additionally supports educational and exploratory workflows for users with limited programming experience.

Rather than replacing existing pathway-oriented or graph-analytic platforms, BiGGViz was designed to occupy a complementary niche within COBRA-centered systems biology workflows. The framework emphasizes rapid exploratory interrogation of stoichiometric network topology directly from COBRA-compatible model structures while maintaining tight integration with MATLAB-based metabolic modeling pipelines. By supporting dynamic annotation overlays, topology-aware filtering, and both reaction-centric and metabolite-centric network representations, BiGGViz facilitates iterative interpretation of

metabolic reconstructions across exploratory, analytical, and communication-oriented workflows.

Materials and Methods

Software overview and workflow

BiGGViz is implemented as a MATLAB-based visualization framework for exploratory analysis of COBRA-compatible metabolic models. Starting from COBRA-compatible model structures, the framework constructs reaction-centric and metabolite-centric network representations directly from the underlying stoichiometric matrix S . These derived networks may subsequently be explored interactively through the BiGGViz graphical interface or exported for downstream visualization and graph-analysis workflows.

A central design principle of BiGGViz is the separation of stoichiometric network topology from analysis-derived annotation overlays. The underlying BiGG/COBRA model defines the metabolic network structure and provides core attributes including metabolite identifiers, reaction identifiers, subsystem labels, reversibility information, gene associations, and reaction formulae. Optional annotation tables allow users to append additional reaction- or metabolite-level metadata without modifying the underlying COBRA model representation. These overlays may include pathway classifications, flux balance analysis (FBA) outputs, flux variability analysis (FVA) summaries, reaction knockout results, differential flux measures, statistical scores, node scaling variables, or other user-defined attributes generated externally or directly within the COBRA Toolbox. Because annotation overlays are incorporated dynamically during visualization, BiGGViz supports iterative and reproducible exploratory workflows linking COBRA-derived analyses to network-centric interpretation.

Figure 1 summarizes the overall BiGGViz framework and analysis workflow. COBRA-compatible metabolic reconstructions and optional annotation tables are first transformed into reaction-centric and metabolite-centric network representations derived directly from the stoichiometric matrix. These networks are subsequently explored within the BiGGViz environment using **topology-aware filtering strategies including degree-based removal of highly connected nodes and minimum connected-component size thresholds**, pathway highlighting, neighborhood expansion, subnetwork extraction, and dynamic annotation overlays. Resulting network views may then be exported as publication-quality static figures, interactive HTML visualizations, or Cytoscape-compatible node and edge tables for downstream analysis and graph-based workflows.

The BiGGViz graphical interface was implemented using MATLAB App Designer and provides an interactive environment for exploratory analysis of genome-scale metabolic networks. Users may load COBRA-compatible models from MAT or SBML/XML files together with optional annotation tables in CSV or XLSX format. Annotation tables are standardized through a parsing routine that aligns reaction or metabolite identifiers with the underlying model structure prior to visualization.

BiGGViz supports both reaction-centric and metabolite-centric exploration modes. Multiple visualization encodings are available, including coloring by pathway, subsystem, reversibility, gene association status, metabolite class, compartment, or

flux-derived quantities. Node sizes may additionally be mapped to network degree, flux magnitude, knockout-derived scores, or other user-defined quantitative variables, allowing biological or topological characteristics to be emphasized within the network representation. **A configurable Label By control further allows on-graph node labels to be mapped to selected model- or annotation-table fields, including identifiers, names, pathway or compartment annotations, reaction formulae, degree values, flux summaries, and metabolite classes.**

To improve the interpretability of large genome-scale reconstructions, BiGGViz incorporates several topology-aware filtering strategies. Users may suppress highly connected currency metabolites, restrict visualization to subnetworks that meet component-size thresholds, or remove dense hub structures via degree-based filtering. These operations reduce visual clutter and facilitate pathway-level interpretation within highly connected metabolic networks.

The framework additionally supports neighborhood-based and pathway-guided exploration workflows. Users may perform hop-based neighborhood expansion around selected metabolites, reactions, or pathways, enabling localized inspection of biochemical connectivity and pathway cross-talk within larger reconstructions. Interactive highlighting, pathway filtering, and search functionality further support rapid navigation of dense genome-scale networks.

BiGGViz was designed to operate naturally alongside existing COBRA analysis workflows. Outputs generated through COBRA Toolbox analyses, including FBA-predicted fluxes, FVA-derived flux ranges, reaction knockout results, pathway assignments, and user-generated quantitative summaries, can be directly appended to annotation tables and rendered dynamically within the network visualization layer. Numerical attributes may be mapped to **node color or node size**, while categorical attributes may be used for pathway grouping and interactive filtering. This integration enables users to contextualize COBRA-derived analytical results within the broader topology of the metabolic network.

At any point, the current network view can be exported directly from the interface as interactive HTML viewers, publication-ready PNG or PDF figures, or Cytoscape-compatible node and edge tables for downstream graph analysis. These export capabilities facilitate transitions between exploratory visualization in MATLAB and external network analysis or presentation workflows.

BiGGViz is implemented in MATLAB and depends on the COBRA Toolbox for model representation, input/output routines, and optional flux analysis. The framework operates on COBRA-compatible model structures, including BiGG models distributed in MAT or SBML formats and user-generated reconstructions loaded through standard COBRA functions such as `readCbModel` and `loadBiggModel`. Although development and testing were primarily conducted on Windows systems, the framework itself is platform-independent, aside from the availability of MATLAB and COBRA.

For users preferring code-driven workflows, BiGGViz also provides a lightweight, scriptable wrapper, `exportBiGGViz`, that invokes the same network-construction routines used by the graphical application. Given a COBRA-compatible model and an optional annotation table, `exportBiGGViz` generates interactive reaction and metabolite network viewers directly from

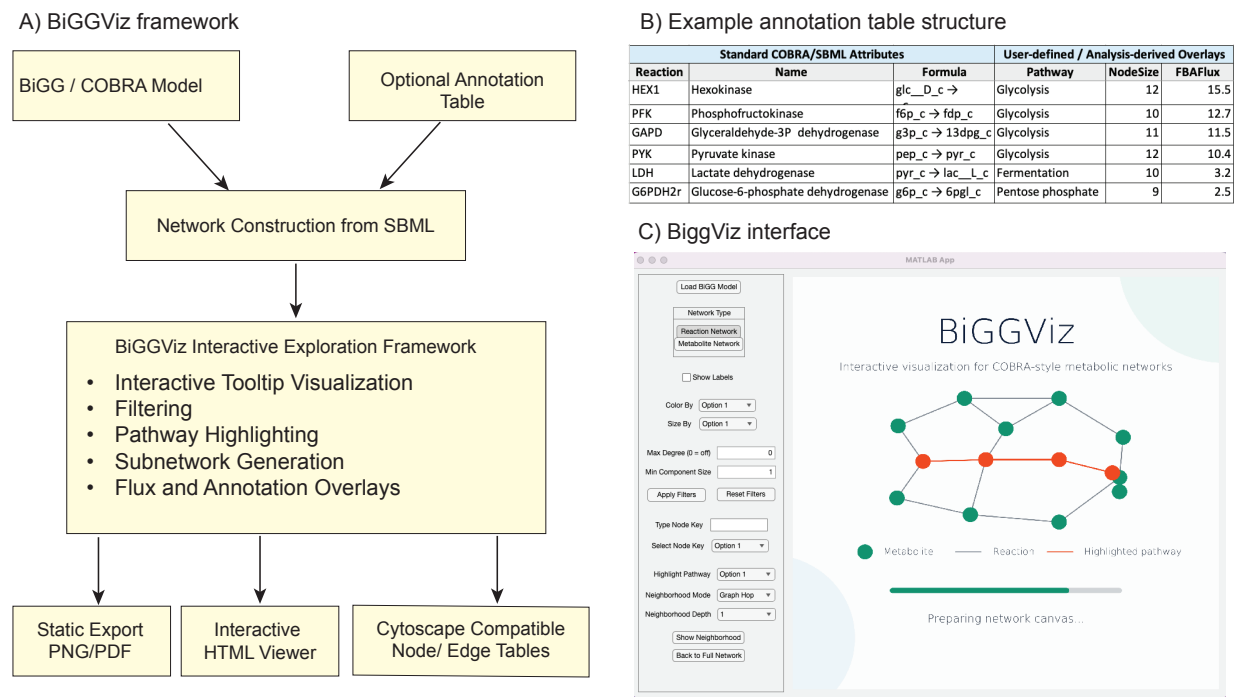


Figure 1 Overview of the BiGGViz framework and analysis workflow. (A) COBRA-compatible BiGG metabolic models are transformed into reaction-centric and metabolite-centric network representations derived directly from the stoichiometric matrix. Optional annotation tables provide an extensible mechanism for integrating user-defined and analysis-derived overlays without modifying the underlying COBRA model structure. These overlays may include pathway annotations, FBA-predicted fluxes, FVA summaries, reaction knockout results, differential flux measures, statistical scores, node scaling variables, or other metadata generated through COBRA Toolbox workflows or external analyses. BiGGViz supports interactive network interrogation through topology-aware filtering strategies, pathway highlighting, neighborhood expansion, subnetwork generation, and dynamic annotation overlays. Resulting visualizations may be exported as publication-quality static figures, interactive HTML viewers, or Cytoscape-compatible node and edge tables for downstream analysis. (B) Example annotation table structure integrating standard COBRA/SBML attributes with user-defined and analysis-derived overlays. (C) BiGGViz graphical interface for interactive exploration of genome-scale metabolic networks.

MATLAB scripts, enabling batch processing of multiple reconstructions and integration into reproducible COBRA analysis pipelines without manual interaction through the GUI.

BiGGViz was tested on more than 50 curated BiGG reconstructions spanning a broad range of organism types, network sizes, and model complexity, ranging from the compact *E. coli* core model (~72 metabolites, ~95 reactions, 137 genes) to large human-scale reconstructions including *iAB-RBC-283* (342 metabolites, 469 reactions, 346 genes) and Recon3D (5835 metabolites, 10 600 reactions, 2248 genes). Models distributed in both MAT and SBML formats were successfully imported and visualized within the framework.

Results

Exploratory visualization of genome-scale metabolic networks

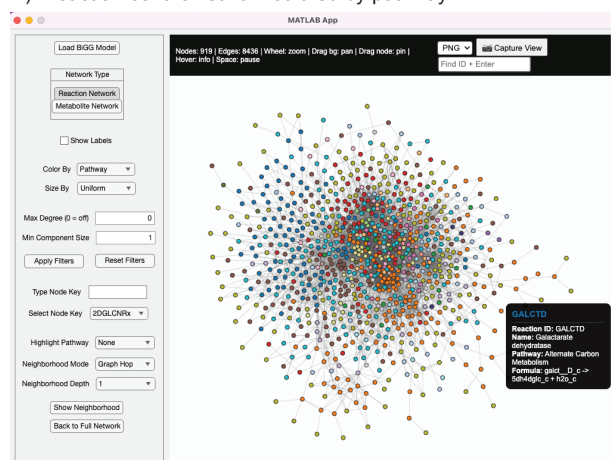
To demonstrate representative BiGGViz workflows, we applied the framework to the human red blood cell reconstruction *iAB-RBC-283* obtained from the BiGG Models database (King et al., 2016; Bordbar et al., 2011). The model was imported directly from the COBRA-compatible MAT format, after which BiGGViz

automatically generated both reaction-centric and metabolite-centric network representations from the underlying stoichiometric matrix without requiring additional graph-construction steps.

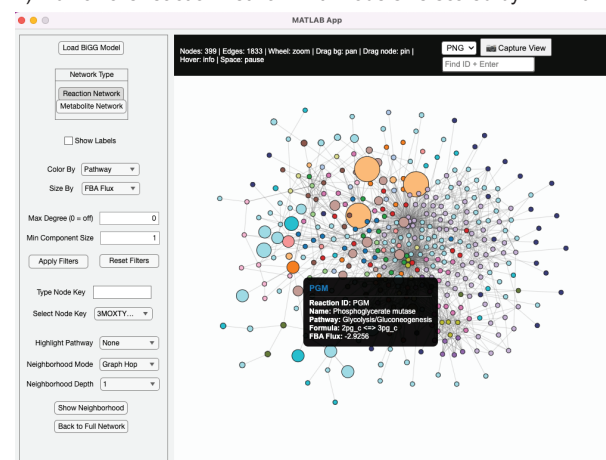
Figure 2 illustrates representative exploratory and analysis-integrated workflows supported within BiGGViz using reaction-centric network representations derived from the *iAB-RBC-283* reconstruction. Although the examples shown here emphasize reaction-reaction connectivity, the same framework additionally supports metabolite-centric network generation and exploration directly from the same underlying COBRA model structure. Unlike manually curated pathway maps, BiGGViz-derived network representations can be dynamically re-encoded using pathway-level, topological, or COBRA-derived analytical attributes without requiring manual reconstruction or redrawing of metabolic layouts. Because the framework operates directly on COBRA-compatible model structures, network topology and downstream analytical overlays remain tightly integrated throughout the exploratory workflow.

Figure 2A shows a reaction-centric network representation colored according to metabolic pathway assignments. This visualization provides a compact overview of pathway organization and the global network structure while preserving relationships among reactions across multiple metabolic subsystems.

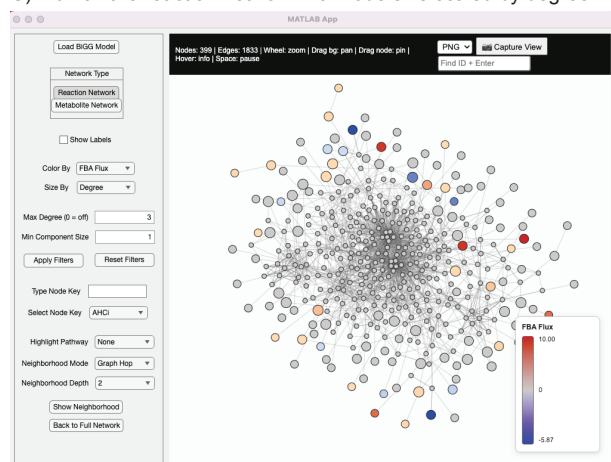
A) Reaction-centric network colored by pathway



B) Flux-aware reaction network with node size scaled by FBA flux



C) Flux-aware reaction network with node size scaled by degree



D) One-hop neighborhood extraction surrounding galactose metabolism

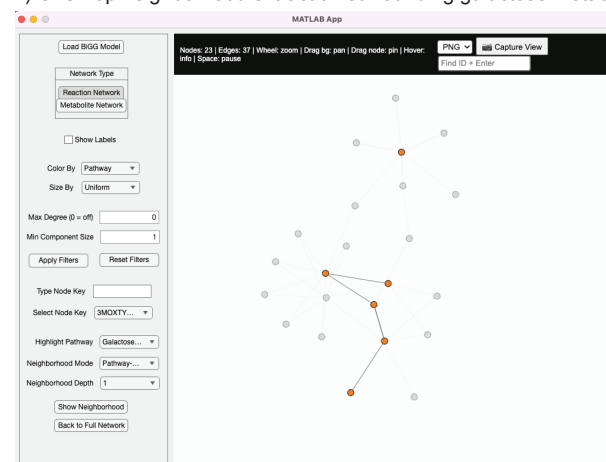


Figure 2 Representative exploratory and analysis-integrated workflows in BiGGViz using the human red blood cell *iAB-RBC-283* reconstruction. (A) Reaction-centric network representation colored according to metabolic pathway assignments, illustrating global pathway organization and network topology within the reconstruction. (B) Flux-aware reaction-centric network in which node size is scaled according to FBA-predicted flux magnitude, enabling visualization of highly active reactions within the broader metabolic network. (C) Multi-attribute reaction-centric network representation with node size mapped to reaction degree and node color mapped to FBA-derived flux values, simultaneously highlighting network topology and predicted metabolic activity. (D) One-hop neighborhood extraction surrounding the galactose metabolism pathway, demonstrating localized pathway-guided exploration and focused inspection of biochemical connectivity within the larger genome-scale reconstruction.

Interactive tooltip overlays further allow users to inspect reaction identifiers, pathway assignments, formulae, and additional annotation metadata directly within the graphical environment.

BiGGViz also supports the direct integration of COBRA-derived analyses via extensible annotation overlays. In Figure 2B, node size is mapped to FBA-predicted flux magnitude, enabling rapid identification of highly active reactions within the metabolic network. These overlays are appended dynamically via annotation tables aligned with COBRA reaction identifiers, demonstrating how outputs from standard COBRA workflows can be incorporated directly into the network visualization layer without modifying the underlying model structure. Flux values may be supplied through user-provided annotation tables aligned with COBRA reaction identifiers; when they are not provided, BiGGViz can compute FBA internally through the COBRA Toolbox and map the resulting flux values onto the

network. Interactive manipulation of network layouts is also supported within the BiGGViz interface. Users may dynamically zoom and pan across large genome-scale networks, manually reposition nodes via drag-and-drop, and temporarily pause force-directed layout updates to stabilize network organization during exploratory analysis or figure generation. These capabilities facilitate detailed inspection of densely connected subnetworks and allow users to refine local network layouts for visualization and interpretation.

More complex multi-attribute encodings are also supported. Figure 2C illustrates a flux-aware reaction-centric network in which node size is mapped to reaction degree while node color represents FBA-derived flux values. This combined representation emphasizes both global network topology and predicted

metabolic activity, allowing highly connected, hub-like reactions and flux-dominant regions to be interpreted within a unified network context.

Beyond global network visualization, BiGGViz supports localized exploratory analysis through reaction-, metabolite-, and pathway-guided subnetwork extraction. Figure 2D demonstrates one-hop neighborhood extraction surrounding the galactose metabolism pathway. This functionality enables focused examination of pathway-local connectivity and biochemical relationships while preserving connections to the broader genome-scale reconstruction. Similar neighborhood expansion workflows may be applied to metabolites, reactions, pathways, or user-defined subnetworks, facilitating targeted investigation of pathway cross-talk and local network organization.

A central feature of the BiGGViz framework is the integration of user-defined and analysis-derived annotation overlays through external annotation tables aligned to COBRA metabolite or reaction identifiers. These overlays may include pathway classifications, reaction annotations, FBA-predicted fluxes, FVA summaries, knockout results, differential flux measures, statistical scores, node scaling variables, gene associations, or other externally generated metadata. Imported overlays are incorporated dynamically during visualization and may subsequently drive node coloring, node scaling, filtering, highlighting, and subnetwork extraction workflows. This flexible annotation architecture allows users to iteratively integrate COBRA-derived analyses with network-centric interpretation while preserving the integrity of the original stoichiometric reconstruction.

Although Figure 2 illustrates the relatively compact *iAB-RBC-283* reconstruction, identical workflows were successfully applied to substantially larger genome-scale reconstructions, including human-scale BiGG models containing thousands of metabolites and reactions. Exported Cytoscape-compatible node and edge tables preserved both intrinsic COBRA annotations and user-defined overlay variables, supporting downstream graph-analysis workflows outside the MATLAB environment.

Collectively, these examples demonstrate how BiGGViz integrates topology-aware visualization, pathway-level organization, and COBRA-derived analytical overlays within a unified framework for exploratory interpretation of genome-scale metabolic reconstructions. By supporting both reaction-centric and metabolite-centric network representations, the framework enables complementary perspectives on biochemical organization, pathway connectivity, and metabolic state within large-scale stoichiometric models.

Discussion

Genome-scale metabolic reconstructions have become increasingly central to systems biology and constraint-based modeling workflows, yet the interpretation of these large, densely connected biochemical systems remains challenging. Although COBRA-based analyses such as flux balance analysis (FBA), flux variability analysis (FVA), and reaction knockout studies provide powerful quantitative insights into metabolic behavior, interpreting these outputs often requires additional network-level context (Orth et al., 2010; Gudmundsson and Thiele, 2010; Heirendt et al., 2019). In particular, understanding how flux changes, pathway perturbations, or localized subnetworks relate to the broader topology of the metabolic system can be difficult

within large genome-scale reconstructions containing thousands of metabolites and reactions.

BiGGViz was developed as an exploratory visualization framework for COBRA-compatible metabolic models in MATLAB. By directly deriving reaction-centric and metabolite-centric network representations from stoichiometric reconstructions, the framework enables interactive exploration of metabolic connectivity without requiring manual pathway curation or external graph-construction workflows. Because BiGGViz operates directly on standard COBRA model structures, users may iteratively transition between COBRA-based analysis and network-centric visualization within a unified computational environment.

A central feature of the framework is the integration of biologically meaningful annotations and analysis-derived overlays directly into network visualizations. Through flexible annotation tables aligned with COBRA metabolite or reaction identifiers, users can dynamically append pathway labels, FBA-derived fluxes, FVA summaries, knockout results, statistical scores, node-scaling variables, and external analysis results without altering the underlying stoichiometric reconstruction. These overlays may subsequently drive node coloring, node scaling, filtering, highlighting, and subnetwork extraction workflows, enabling iterative interpretation of COBRA-derived analytical results within the broader topology of the metabolic network.

The framework additionally emphasizes exploratory interaction with large genome-scale networks. Interactive tooltip-based inspection enables rapid access to reaction- and metabolite-level metadata, including identifiers, formulae, pathway annotations, and user-defined analytical overlays. Dynamic zooming, panning, drag-based node repositioning, and force-directed layout controls further support detailed inspection of dense subnetworks and local pathway organization. Together, these capabilities facilitate interpretation of pathway cross-talk, neighborhood structure, and network connectivity patterns within highly interconnected metabolic systems.

Beyond visualization, the network representations generated within BiGGViz may also support downstream graph-theoretic and network-analytic workflows. Exported reaction-reaction and metabolite-metabolite networks can be integrated into external graph-analysis environments to estimate network measures such as degree, betweenness, and centrality, as well as to perform community detection, subnetwork analysis, and pathway-level network interrogation. Cytoscape-compatible node and edge exports additionally facilitate interoperability with broader graph-analysis ecosystems and downstream systems biology workflows (Shannon et al., 2003).

Existing metabolic visualization platforms often emphasize different aspects of systems biology workflows. Escher provides highly effective curated pathway visualizations and pathway-map construction, particularly for presentation and pathway-focused interpretation, but is less oriented toward exploratory whole-network graph analysis directly derived from genome-scale stoichiometric reconstructions (King et al., 2015). Conversely, Cytoscape offers a powerful and extensible ecosystem for graph analytics and network visualization, although workflows typically require export into external environments and additional preprocessing steps (Shannon et al., 2003). Other platforms such as MetExploreViz and related network-analysis tools similarly emphasize distinct visualization or analytical objectives (Chazalviel et al., 2018; Gavai et al., 2015; Konganti et al., 2013; Posma

et al., 2014). BiGGViz was designed to complement, rather than replace, these tools by providing a lightweight exploratory layer tightly integrated with COBRA-compatible modeling workflows in MATLAB.

Several limitations should be acknowledged. Genome-scale metabolic reconstructions remain inherently dense and visually complex, and highly connected currency metabolites may still complicate interpretation despite topology-aware filtering strategies. In addition, BiGGViz currently emphasizes exploratory structural visualization and network interrogation rather than advanced graph analytics or dynamic simulation. Although the framework was designed around MATLAB/COBRA interoperability, users operating primarily within Python- or web-based ecosystems may require additional export or integration steps. Future development will focus on expanding interoperability through native Python integration, enhancing multi-omics annotation support, developing comparative and condition-aware visualization workflows, and implementing scalable rendering strategies for increasingly large human-scale reconstructions.

Overall, BiGGViz provides a flexible, exploratory visualization framework for interpreting COBRA-compatible metabolic reconstructions. By integrating topology-aware visualization, pathway-level organization, and analysis-derived overlays into a unified environment that supports both reaction-centric and metabolite-centric exploration, the framework facilitates network-centric interpretation of large-scale metabolic systems in reproducible systems biology workflows.

Conflicts of interest

The authors declare that they have no competing interests.

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Data availability

BiGGViz source code, example datasets, and tutorial materials are freely available through the project GitHub repository at <https://github.com/mogystats/BiGGViz>. Sample COBRA-compatible metabolic models and annotation tables used for demonstration and testing are provided within the repository.

Author contributions statement

R.H.B. and S.-L.Y. jointly conceived and designed the study, developed the BiGGViz framework, conducted the analyses, interpreted the results, and wrote and reviewed the manuscript.

References

- S. A. Becker and B. O. Palsson. Context-specific metabolic networks are consistent with experiments. *PLoS Computational Biology*, 4(5):e1000082, 2008. doi: 10.1371/journal.pcbi.1000082.
- A. Bordbar, N. Jamshidi, and B. O. Palsson. iab-rbc-283: A proteomically derived knowledge-base of erythrocyte metabolism that can be used to simulate its physiological and patho-physiological states. *BMC Systems Biology*, 5(1):110, 2011. doi: 10.1186/1752-0509-5-110.
- M. Chazalviel, C. Frainay, N. Poupin, F. Vinson, B. Merlet, Y. Gloaguen, L. Cottret, and F. Jourdan. Metexploreviz: web component for interactive metabolic network visualization. *Bioinformatics*, 34(2):312–313, 2018.
- A. K. Gavai, F. Supandi, H. Hettling, P. Murrell, J. A. Leunissen, and J. H. Van Beek. Using bioconductor package biggr for metabolic flux estimation based on gene expression changes in brain. *PLoS One*, 10(3):e0119016, 2015.
- S. Gudmundsson and I. Thiele. Computationally efficient flux variability analysis. *BMC bioinformatics*, 11(1):489, 2010.
- L. Heirendt, S. Arreckx, T. Pfau, et al. Creation and analysis of biochemical constraint-based models: the COBRA toolbox v3.0. *Nature Protocols*, 14:639–702, 2019. doi: 10.1038/s41596-018-0098-2. URL <https://doi.org/10.1038/s41596-018-0098-2>.
- Z. A. King, A. Dräger, A. Ebrahim, N. Sonnenschein, N. E. Lewis, and B. O. Palsson. Escher: a web application for building, sharing, and embedding data-rich visualizations of biological pathways. *PLoS computational biology*, 11(8):e1004321, 2015.
- Z. A. King, J. Lu, A. Dräger, P. Miller, S. Federowicz, J. A. Lerman, A. Ebrahim, B. O. Palsson, and N. E. Lewis. Bigg models: A platform for integrating, standardizing and sharing genome-scale models. *Nucleic acids research*, 44(D1):D515–D522, 2016.
- K. Konganti, G. Wang, E. Yang, and J. J. Cai. Sbttoolbox: a matlab toolbox for biological network analysis. *Evolutionary Bioinformatics*, 9:EBO–S12012, 2013.
- J. D. Orth, I. Thiele, and B. Ø. Palsson. What is flux balance analysis? *Nature biotechnology*, 28(3):245–248, 2010.
- J. M. Poma, S. L. Robinette, E. Holmes, and J. K. Nicholson. Metabonetworks, an interactive matlab-based toolbox for creating, customizing and exploring sub-networks from kegg. *Bioinformatics*, 30(6):893–895, 2014.
- J. Schellenberger, R. Que, R. M. Fleming, I. Thiele, J. D. Orth, A. M. Feist, D. C. Zielinski, A. Bordbar, N. E. Lewis, S. Rahmanian, et al. Quantitative prediction of cellular metabolism with constraint-based models: the cobra toolbox v2. 0. *Nature protocols*, 6(9):1290–1307, 2011.
- P. Shannon, A. Markiel, O. Ozier, N. S. Baliga, J. T. Wang, D. Ramage, N. Amin, B. Schwikowski, and T. Ideker. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Research*, 13(11):2498–2504, 2003. doi: 10.1101/gr.1239303.